OSTAD BATCH -6 AI ENGINEERING

MODULE 17

BANGLADESHI TAKA DETECTION

==============================

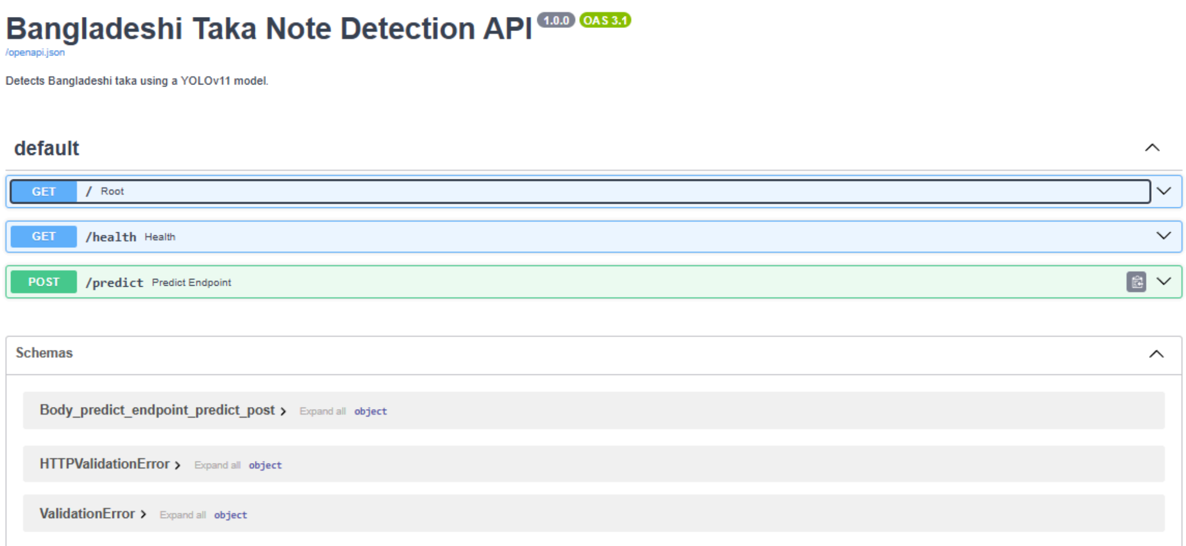
Overview

This project detects the denomination of a Bangladeshi Taka note from an image. It uses a YOLOv11 model for detection and serves it with a FastAPI REST API. The whole app is packed inside Docker so it can run anywhere. When you send an image to the API, it tells you which note it is (like 100 taka), how confident the model is, and the bounding box of the note in the image. Visit https://taka-note-api-953584645385.us-central1.run.app/docs for live demo and https://github.com/unoxidized-copper/taka-note-detection for repository.

INFERENCE


```
Image: test_2_taka.png  (size=(256, 117))
   -> 2_taka      conf=0.9656  bbox=[0.2, 0.6, 256.0, 117.0]

Image: test_5_taka.png  (size=(256, 117))
   -> 5_taka      conf=0.9468  bbox=[1.8, 0.8, 256.0, 117.0]

Image: test_10_taka.png  (size=(256, 117))
   -> 10_taka     conf=0.9551  bbox=[0.3, 0.6, 254.4, 117.0]

Image: test_20_taka.png  (size=(256, 117))
   -> 20_taka     conf=0.9706  bbox=[1.2, 0.8, 255.7, 117.0]

Image: test_50_taka.png  (size=(256, 117))
   -> 50_taka     conf=0.9779  bbox=[0.2, 0.6, 255.8, 117.0]

Image: test_100_taka.png  (size=(256, 117))
   -> 100_taka    conf=0.9176  bbox=[0.6, 1.5, 256.0, 117.0]

Image: test_500_taka.png  (size=(256, 117))
   -> 500_taka    conf=0.9795  bbox=[0.3, 0.8, 256.0, 117.0]

Image: test_1000_taka.png  (size=(256, 117))
   -> 1000_taka   conf=0.9553  bbox=[1.4, 0.9, 255.4, 117.0]
```

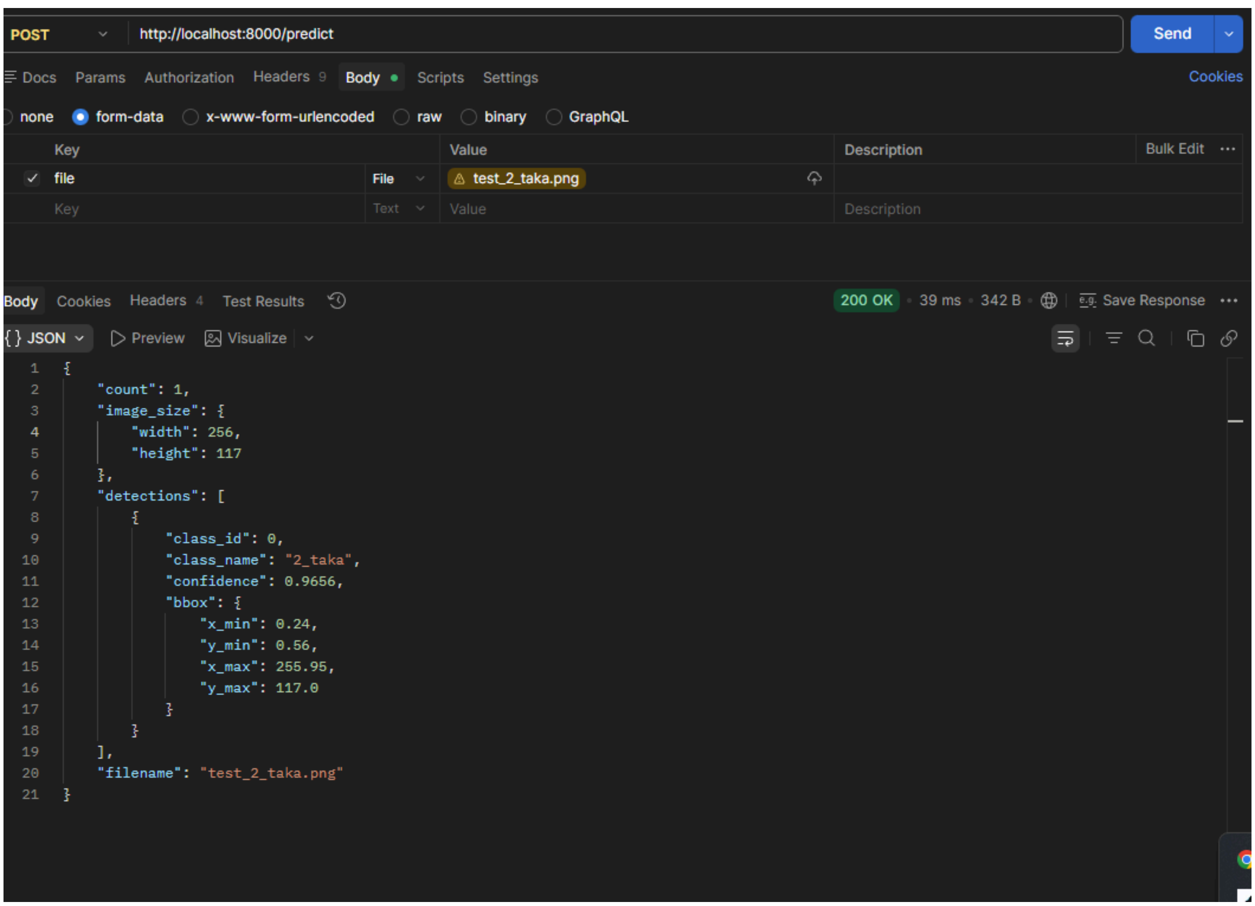
API

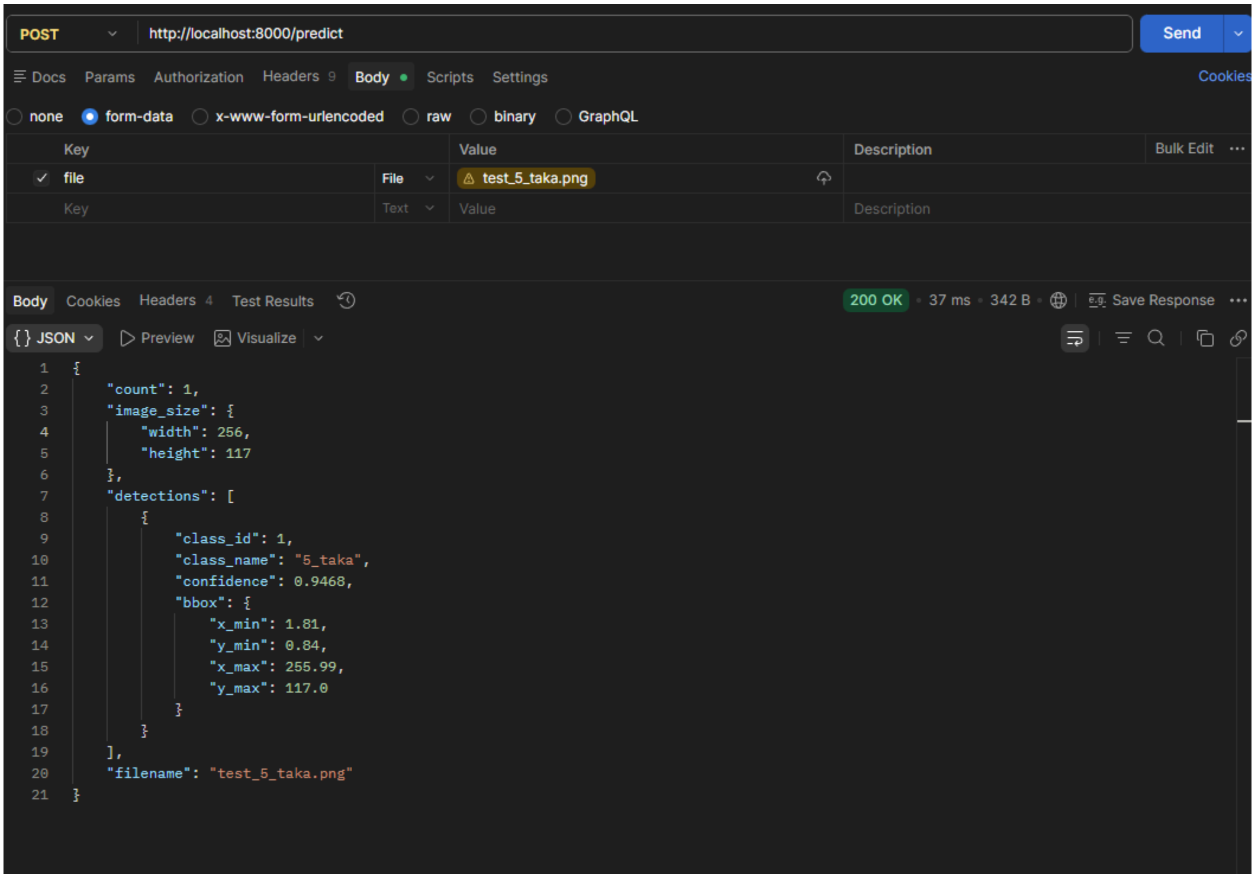

Bangladeshi Taka Note Detection API
1.0.0
OAS 3.1
/openapi.json
Detects Bangladeshi taka using a YOLOv11 model.
default
GET
/ Root
GET
/health Health
POST
/predict Predict Endpoint
Schemas
Body_predict_endpoint_predict_post
Expand all
object
HTTPValidationError
Expand all
object
ValidationError
Expand all
object

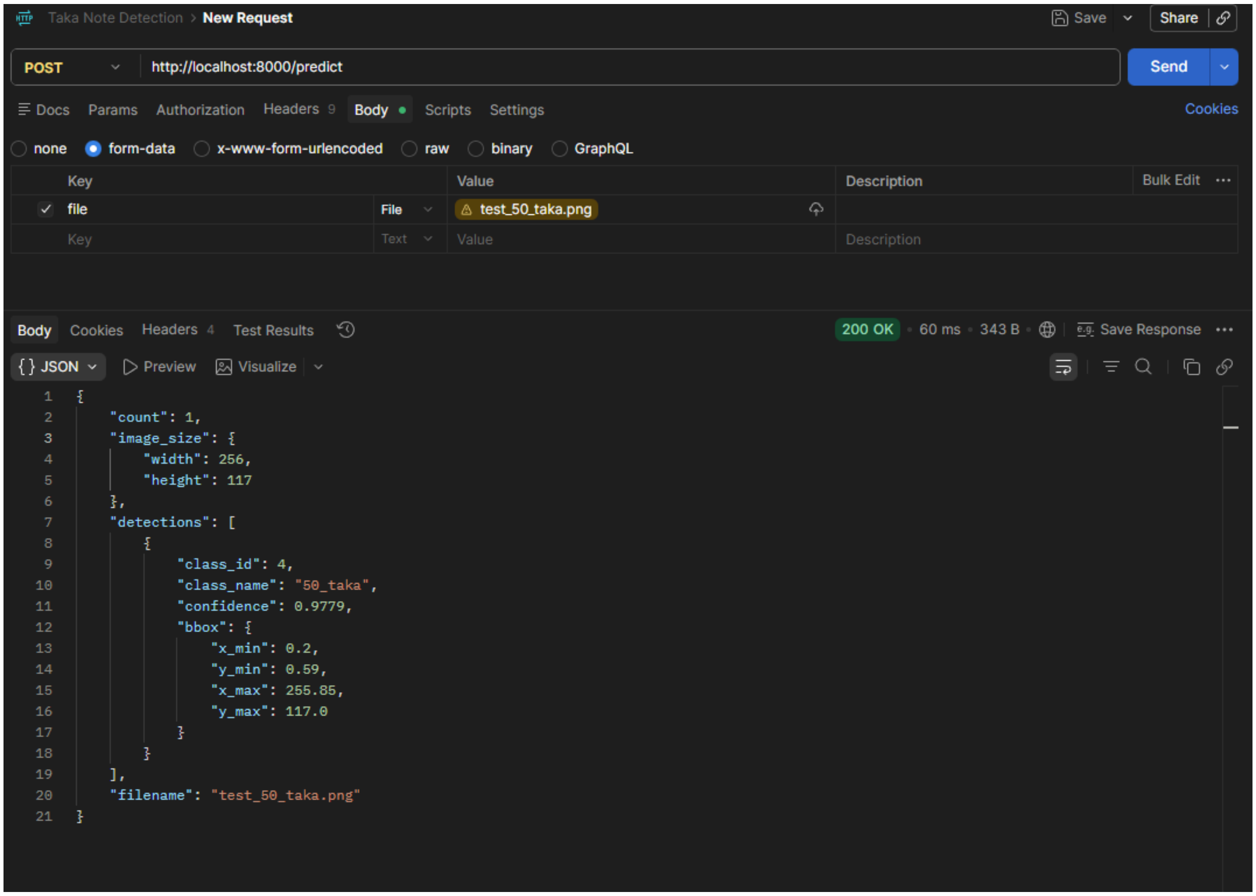
TESTING

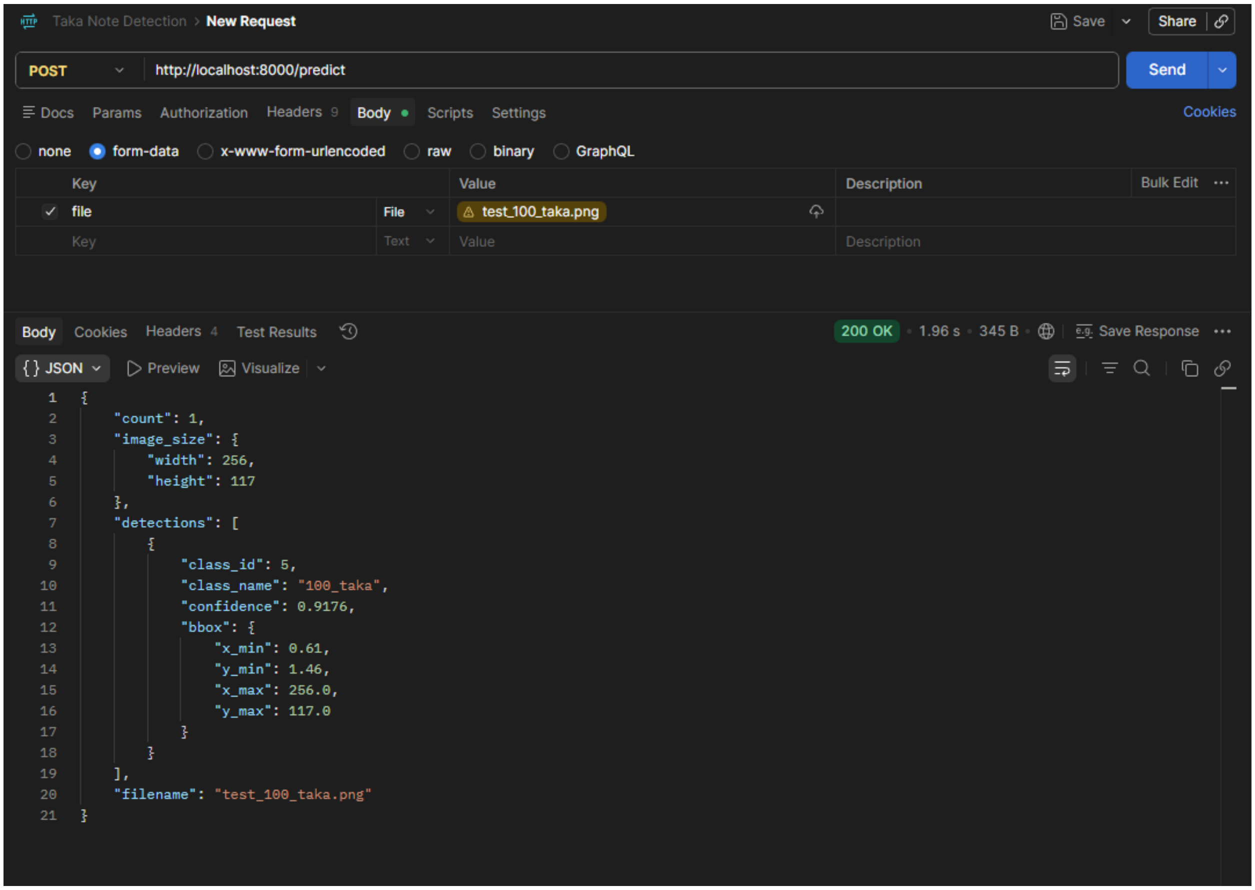
Prediction accuracy discussion

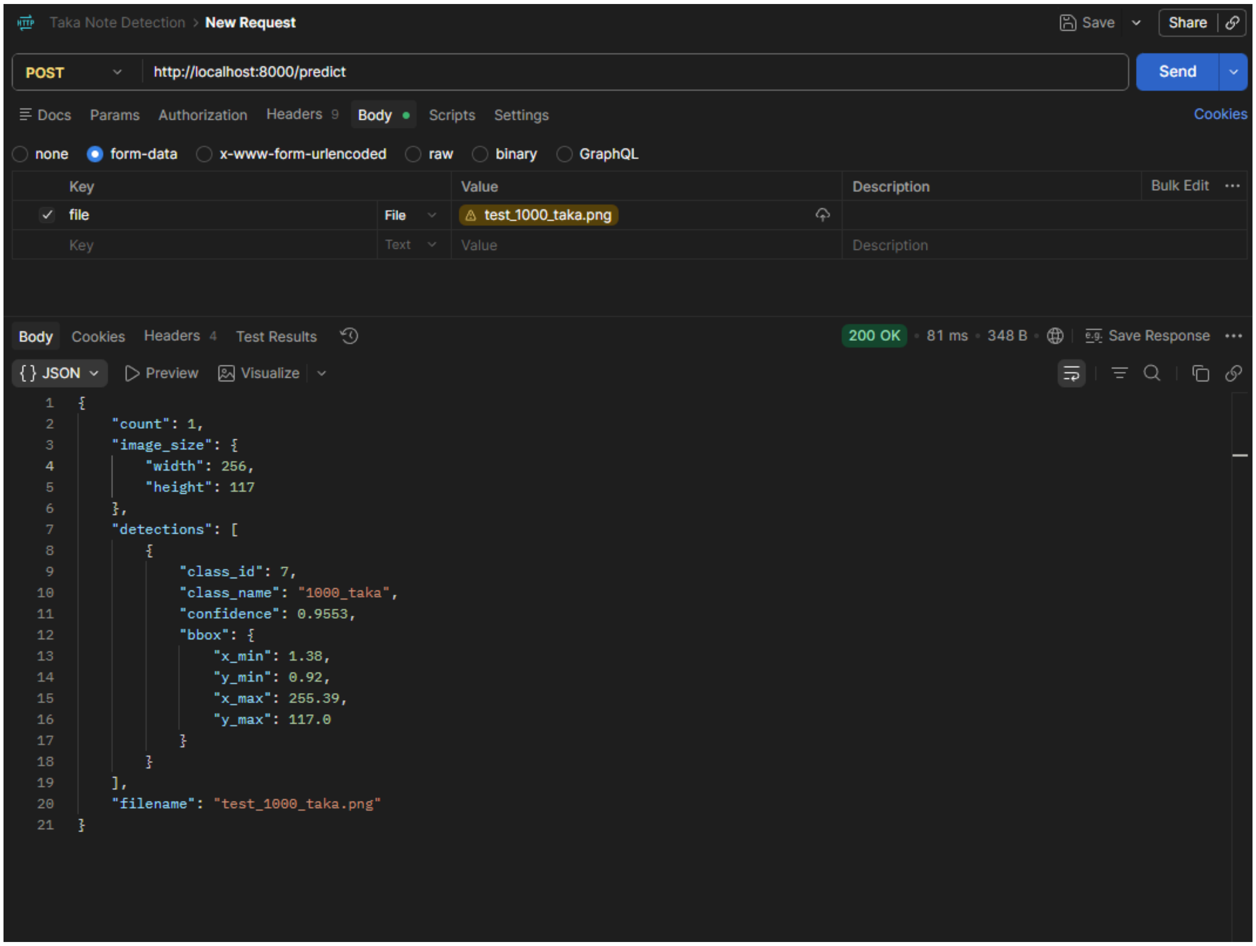
I tested the API with 8 different images (2, 5, 10, 20, 50, 100, 500 and 1000 taka). The model predicted every note correctly. The confidence scores were high, between 0.91 and 0.98. The bounding boxes covered almost the full image, which is expected because in this dataset each image is a single note that fills the whole frame. On the validation set of 600 images the model got precision 0.999, recall 1.000 and mAP@50 0.995, so the results from the API match the validation performance. The accuracy is very high mainly because one clear note per image with no background clutter.

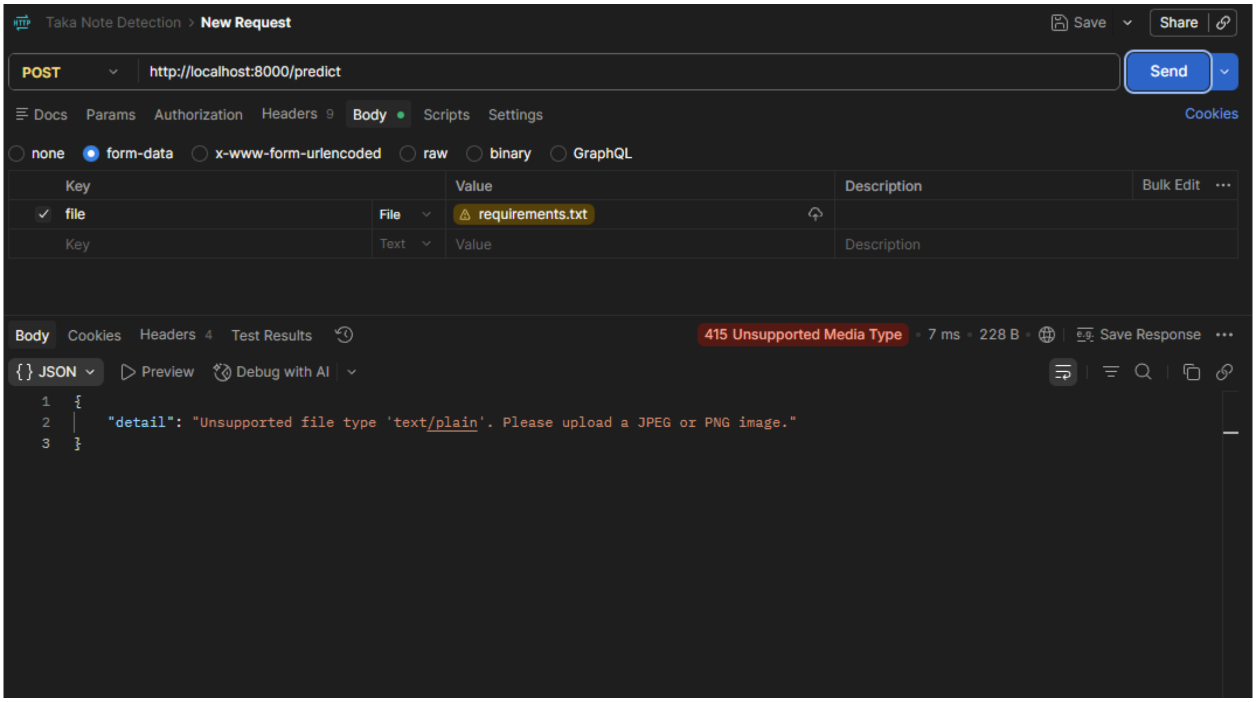


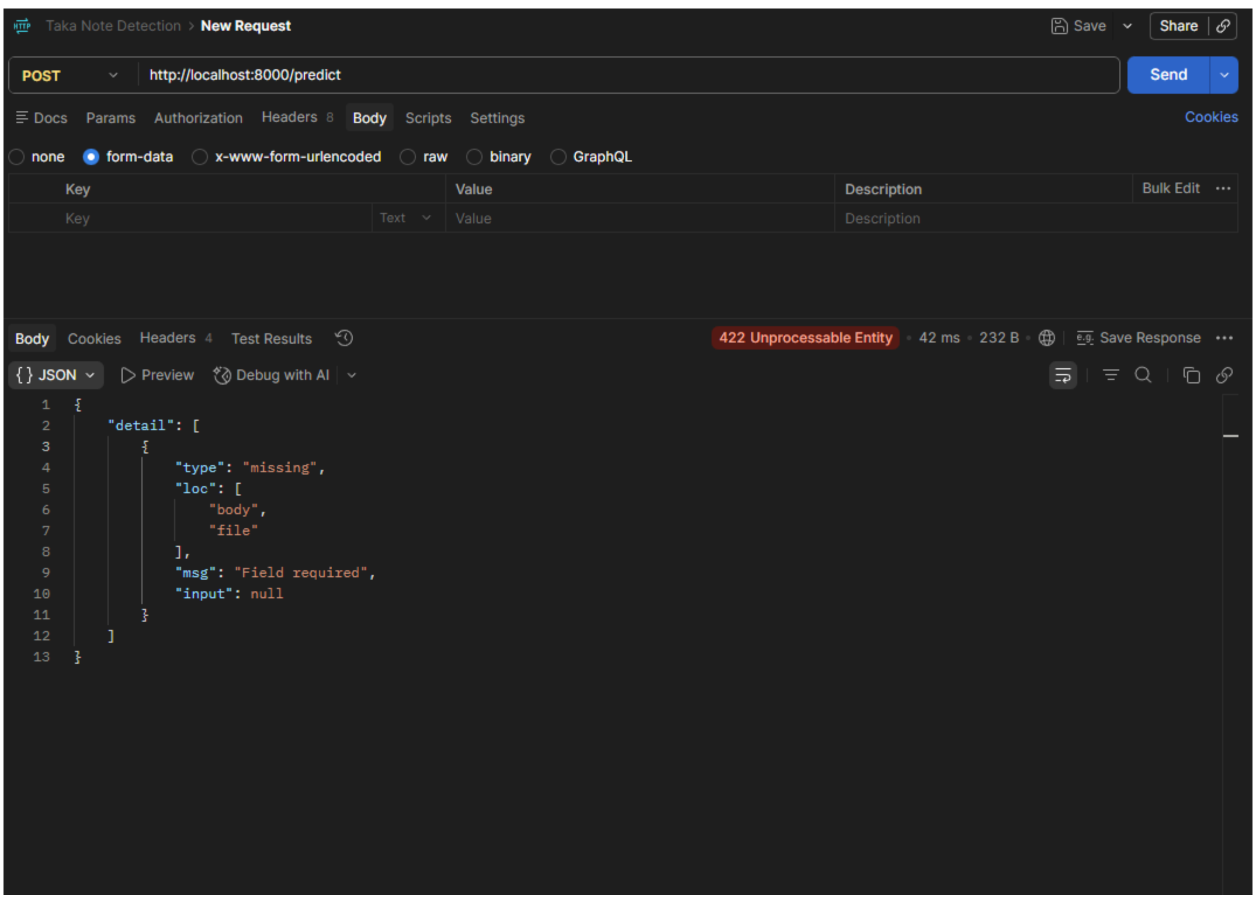

POST
http://localhost:8000/predict
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
Key Value Description Bulk Edit
file File test_5_taka.png
Key Text Value Description
Body Cookies Headers 4 Test Results
200 OK 37 ms 342 B Save Response
{} JSON Preview Visualize
{
"count": 1,
"image_size": {
"width": 256,
"height": 117
},
"detections": [
{
"class_id": 1,
"class_name": "5_taka",
"confidence": 0.9468,
"bbox": {
"x_min": 1.81,
"y_min": 0.84,
"x_max": 255.99,
"y_max": 117.0
}
}
],
"filename": "test_5_taka.png"
}

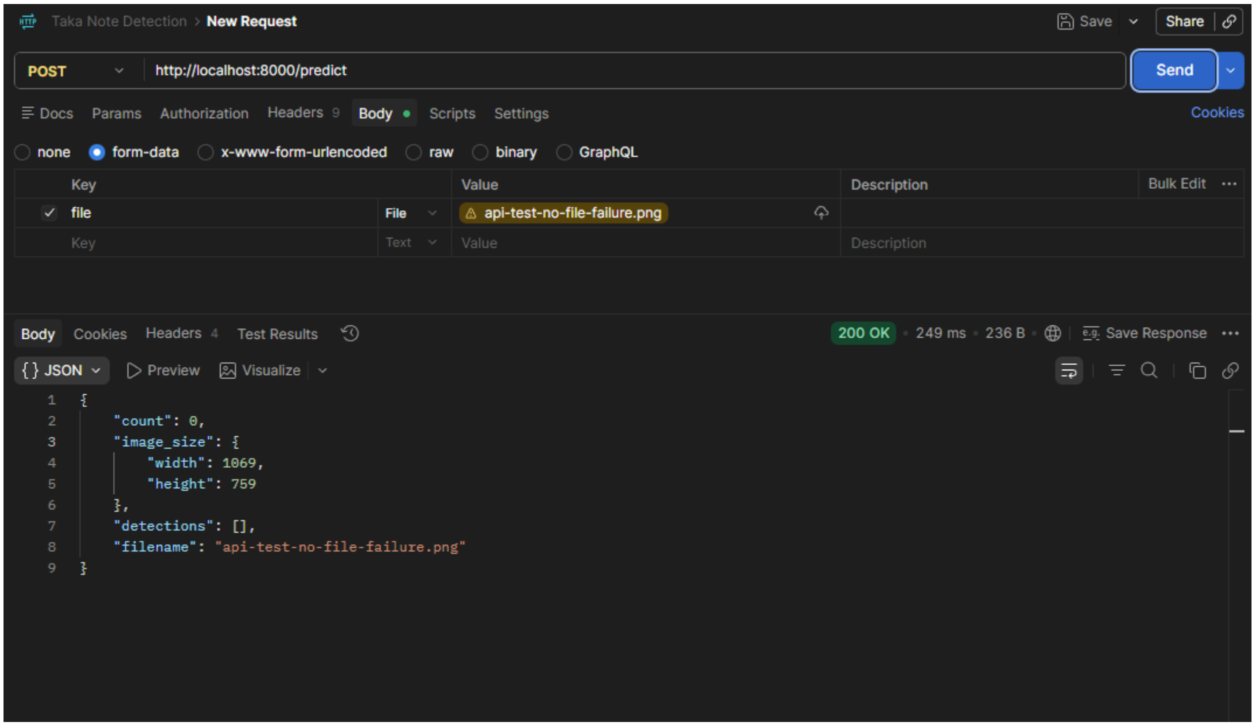

Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
test_50_taka.png
Key
Text
Value
Description
Body
Cookies
Headers 4
Test Results
200 OK
60 ms
343 B
Save Response
{} JSON
Preview
Visualize
{
"count": 1,
"image_size": {
"width": 256,
"height": 117
},
"detections": [
{
"class_id": 4,
"class_name": "50_taka",
"confidence": 0.9779,
"bbox": {
"x_min": 0.2,
"y_min": 0.59,
"x_max": 255.85,
"y_max": 117.0
}
}
],
"filename": "test_50_taka.png"
}


Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
test_100_taka.png
Key
Text
Value
Description
Body
Cookies
Headers 4
Test Results
200 OK
1.96 s
345 B
Save Response
{ } JSON
Preview
Visualize
{
"count": 1,
"image_size": {
"width": 256,
"height": 117
},
"detections": [
{
"class_id": 5,
"class_name": "100_taka",
"confidence": 0.9176,
"bbox": {
"x_min": 0.61,
"y_min": 1.46,
"x_max": 256.0,
"y_max": 117.0
}
}
],
"filename": "test_100_taka.png"
}


Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
test_1000_taka.png
Key
Text
Value
Description
Body
Cookies
Headers 4
Test Results
200 OK
81 ms
348 B
Save Response
{ } JSON
Preview
Visualize
{
"count": 1,
"image_size": {
"width": 256,
"height": 117
},
"detections": [
{
"class_id": 7,
"class_name": "1000_taka",
"confidence": 0.9553,
"bbox": {
"x_min": 1.38,
"y_min": 0.92,
"x_max": 255.39,
"y_max": 117.0
}
}
],
"filename": "test_1000_taka.png"
}


Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
Key Value Description Bulk Edit
file File requirements.txt
Key Text Value Description
Body Cookies Headers 4 Test Results
415 Unsupported Media Type 7 ms 228 B
Save Response
{} JSON Preview Debug with AI
{
"detail": "Unsupported file type 'text/plain'. Please upload a JPEG or PNG image."
}


Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs Params Authorization Headers 8 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
Key Value Description Bulk Edit
Key Text Value Description
Body Cookies Headers 4 Test Results
422 Unprocessable Entity 42 ms 232 B
Save Response
{} JSON Preview Debug with AI
{
"detail": [
{
"type": "missing",
"loc": [
"body",
"file"
],
"msg": "Field required",
"input": null
}
]
}



Taka Note Detection > New Request
Save
Share
POST
http://localhost:8000/predict
Send
Docs Params Authorization Headers 9 Body Scripts Settings
Cookies
none form-data x-www-form-urlencoded raw binary GraphQL
Key
Value
Description
Bulk Edit
file
File
api-test-no-file-failure.png
Key
Text
Value
Description
Body Cookies Headers 4 Test Results
200 OK
249 ms
236 B
Save Response
{ } JSON
Preview
Visualize
{
"count": 0,
"image_size": {
"width": 1069,
"height": 759
},
"detections": [],
"filename": "api-test-no-file-failure.png"
}

DOCKER RUN

```
PS C:\Users\BS1123\StuffDoneHere\CodingStuff\pythonStuff\taka-note-detection> docker ps
CONTAINER ID   IMAGE                         COMMAND                  CREATED         STATUS                 PORTS                                       NAMES
21f5edd9100b   taka-note-api                 "uvicorn app.main:ap…"   5 minutes ago   Up 5 minutes           0.0.0.0:8000->8000/tcp                      taka-api
a1fae4f8358a   quay.io/keycloak/keycloak:26.0 "/opt/keycloak/bin/k…"  6 days ago      Up 3 days              8443/tcp, 9000/tcp, 0.0.0.0:8081->8080/tcp  echo-tracker-keycloak
a0f5d2b099a4   postgres:16-alpine            "docker-entrypoint.s…"   6 days ago      Up 3 days (healthy)    0.0.0.0:5433->5432/tcp                      echo-tracker-postgres
```

DEPLOYMENT TO GCP

Deployment steps explanation

I deployed the Dockerized app on Google Cloud Run.

Steps:

(1) Created a Google Cloud project and enabled billing.

(2) Enabled the Cloud Run, Cloud Build and Artifact Registry APIs.

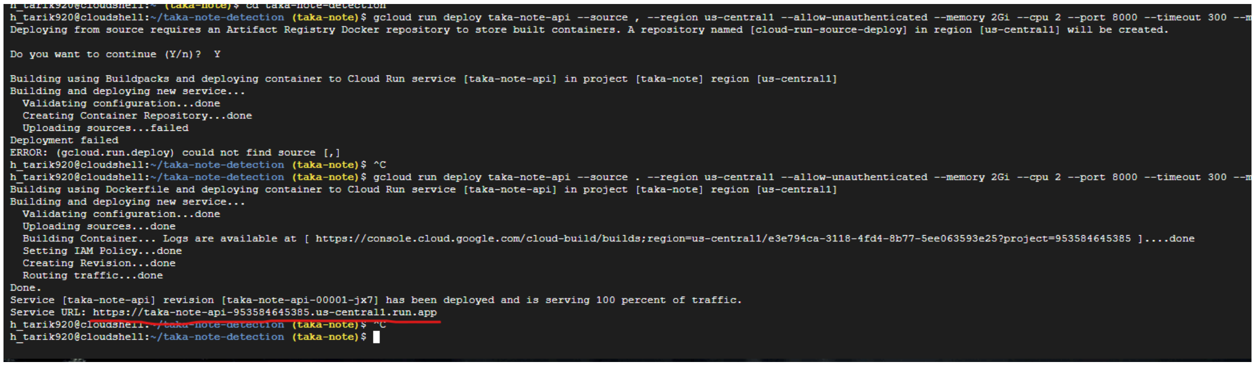
(3) Opened Cloud Shell and cloned my GitHub repo.

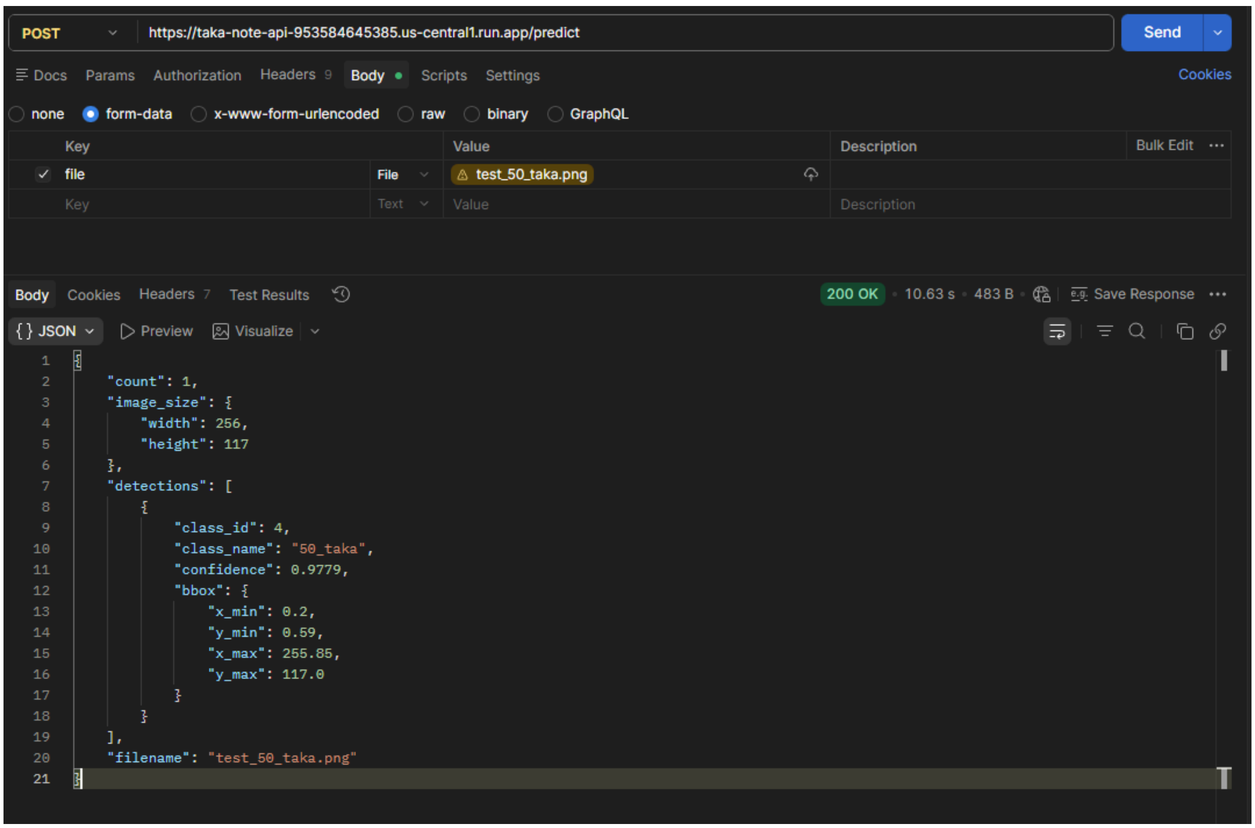
(4) Ran deployment commands specifying port, CPU [google search and AI]

Cloud Build automatically built my Dockerfile and deployed the container.

(5) Cloud Run gave a public HTTPS URL

The service scales to zero when idle so it stays within the free tier.


h_tarik920@cloudshell:~/taka-note-detection (taka-note)$ gcloud run deploy taka-note-api --source , --region us-central1 --allow-unauthenticated --memory 2Gi --cpu 2 --port 8000 --timeout 300 --m
Deploying from source requires an Artifact Registry Docker repository to store built containers. A repository named [cloud-run-source-deploy] in region [us-central1] will be created.
Do you want to continue (Y/n)? Y
Building using Buildpacks and deploying container to Cloud Run service [taka-note-api] in project [taka-note] region [us-central1]
Building and deploying new service...
Validating configuration...done
Creating Container Repository...done
Uploading sources...failed
Deployment failed
ERROR: (gcloud.run.deploy) could not find source [,]
h_tarik920@cloudshell:~/taka-note-detection (taka-note)$ ^C
h_tarik920@cloudshell:~/taka-note-detection (taka-note)$ gcloud run deploy taka-note-api --source . --region us-central1 --allow-unauthenticated --memory 2Gi --cpu 2 --port 8000 --timeout 300 --m
Building using Dockerfile and deploying container to Cloud Run service [taka-note-api] in project [taka-note] region [us-central1]
Building and deploying new service...
Validating configuration...done
Uploading sources...done
Building Container... Logs are available at [https://console.cloud.google.com/cloud-build/builds;region=us-central1/e3e794ca-3118-4fd4-8b77-5ee063593e25?project=953584645385]....done
Setting IAM Policy...done
Creating Revision...done
Routing traffic...done
Done.
Service [taka-note-api] revision [taka-note-api-00001-jx7] has been deployed and is serving 100 percent of traffic.
Service URL: https://taka-note-api-953584645385.us-central1.run.app
h_tarik920@cloudshell:~/taka-note-detection (taka-note)$ ^C
h_tarik920@cloudshell:~/taka-note-detection (taka-note)$


POST
https://taka-note-api-953584645385.us-central1.run.app/predict
Send
Docs
Params
Authorization
Headers 9
Body
Scripts
Settings
Cookies
none
form-data
x-www-form-urlencoded
raw
binary
GraphQL
Key
Value
Description
Bulk Edit
file
File
test_50_taka.png
Key
Text
Value
Description
Body
Cookies
Headers 7
Test Results
200 OK
10.63 s
483 B
Save Response
{} JSON
Preview
Visualize
{
"count": 1,
"image_size": {
"width": 256,
"height": 117
},
"detections": [
{
"class_id": 4,
"class_name": "50_taka",
"confidence": 0.9779,
"bbox": {
"x_min": 0.2,
"y_min": 0.59,
"x_max": 255.85,
"y_max": 117.0
}
}
],
"filename": "test_50_taka.png"
}